

Corrected score, stated first. At window w , score a candidate right direction $v \in \mathbb{S}^{d-1}$ by

$$\text{score}_w(v) := \psi_{\text{old}}(v) + \psi_{\text{new}}(v), \quad (1)$$

where

$$\psi_{\text{old}}(v) := \|Bv\|_2^2 \left(\frac{n_{\text{seen}}}{n} \right)^{-\frac{h_2^{(R)}(v)}{4 \log s}}, \quad (2)$$

$$\psi_{\text{new}}(v) := \|A_w v\|_2^2 \left(\frac{k}{n_w} \right)^{-\frac{h_2^{(w)}(v)}{4 \log k}}. \quad (3)$$

Here ψ_{old} is the corrected contribution from the old sketch and old row memory, while ψ_{new} is the corrected contribution from the current row window. If this is the first window, or if no old sketch/sample is available, set $\psi_{\text{old}}(v) = 0$. The rest of this note defines the symbols and explains the entropy corrections.

Notation in detail. Let $\text{row}(X)$ denote the number of rows of a matrix X . Let

$$A \in \mathbb{R}^{m \times d} \quad (4)$$

be the full data matrix, let A_w denote the current row-window, and let $B \in \mathbb{R}^{r \times d}$ denote the sketch carried from the previous windows. In addition to the sketch, carry a small row sample of old data

$$R \in \mathbb{R}^{s \times d}. \quad (5)$$

Write

$$A_w \in \mathbb{R}^{k \times d}, \quad k = \text{row}(A_w), \quad k > 1, \quad (6)$$

$$n_w := (\# \text{ rows in current window}) + (\# \text{ rows not yet seen after this window}), \quad (7)$$

$$n_{\text{seen}} := \# \text{ raw rows represented by the old sketch/sample before this window}, \quad (8)$$

$$n := \text{total number of rows in the full matrix or stream}, \quad (9)$$

$$v \in \mathbb{S}^{d-1}. \quad (10)$$

Assume

$$s > 1, \quad n_{\text{seen}} > 1, \quad n > 1, \quad (11)$$

whenever an old contribution is present.

For a nonzero response vector y , write

$$h_2 \left(\frac{y}{\|y\|_2} \right) = -\log \left(\frac{\|y\|_4^4}{\|y\|_2^4} \right) = 4 \log \|y\|_2 - 4 \log \|y\|_4. \quad (12)$$

Explanation of the corrected score. The old sketch preserves only quadratic energy, so the old term should not be left as a purely quadratic contribution while the new term receives an entropy correction. The corrected score therefore uses the sketch for old energy and the row sample R for a direction-dependent proxy of old entropy.

For the old rows, define

$$E_{\text{old}}(v) := \|Bv\|_2^2, \quad (13)$$

$$y_R(v) := Rv, \quad (14)$$

$$h_2^{(R)}(v) := -\log \left(\frac{\|Rv\|_4^4}{\|Rv\|_2^4} \right). \quad (15)$$

The sample entropy is not treated as the true old entropy. Instead, use the same normalized-entropy scaling as in the window derivation:

$$\frac{h_2(\text{old full response})}{\log n_{\text{seen}}} \approx \frac{h_2^{(R)}(v)}{\log s}. \quad (16)$$

Thus

$$h_2(\text{old full response}) \approx h_2^{(R)}(v) \frac{\log n_{\text{seen}}}{\log s}. \quad (17)$$

The row sample is used only to approximate the normalized old entropy. The old rows themselves occupy the fraction n_{seen}/n of the full matrix, so the old contribution is

$$\psi_{\text{old}}(v) := E_{\text{old}}(v) \left(\frac{n_{\text{seen}}}{n} \right)^{-\frac{h_2^{(R)}(v)}{4 \log s}}. \quad (18)$$

For the current window, define

$$h_2^{(w)}(v) := h_2 \left(\frac{A_w v}{\|A_w v\|_2} \right) = -\log \left(\frac{\|A_w v\|_4^4}{\|A_w v\|_2^4} \right). \quad (19)$$

The new contribution is

$$\psi_{\text{new}}(v) := \|A_w v\|_2^2 \left(\frac{k}{n_w} \right)^{-\frac{h_2^{(w)}(v)}{4 \log k}}. \quad (20)$$

There is no additional $(k/n_w)^{1/4}$ prefactor in this corrected score.

The final score is

$$\text{score}_w(v) := \psi_{\text{old}}(v) + \psi_{\text{new}}(v). \quad (21)$$

If there is no old sketch/sample, set $\psi_{\text{old}}(v) = 0$.

Algorithm overview. The forget method processes the matrix one row-window at a time.

For the first window, there is no previous sketch and no old row sample, so

$$B = 0, \quad R = \emptyset, \quad \psi_{\text{old}}(v) = 0, \quad (22)$$

and the score is just

$$\text{score}_1(v) = \psi_{\text{new}}(v). \quad (23)$$

Compute retained directions $v_1, \dots, v_r$ sequentially by solving

$$v_j \in \arg \max_{v \in \mathbb{S}^{d-1}, v_i^\top v = 0 \ \forall i < j} \text{score}_1(v). \quad (24)$$

Collecting them gives a retained right basis

$$V_{\text{hat}} = [v_1, \dots, v_r]. \quad (25)$$

With the current carried operator $M = A_1$, form

$$Y = MV_{\text{hat}}, \quad U = \text{orth}(Y), \quad (26)$$

compute the SVD of

$$U^\top M = U_{\text{small}} \Sigma V_{\text{new}}^\top, \quad (27)$$

and carry the sketch

$$B_{\text{next}} = \Sigma V_{\text{new}}^\top. \quad (28)$$

Also update the old-row memory R_{next} from the rows seen so far. The row memory should contain a uniform reservoir component for coverage and, optionally, a small high-response component selected by large responses on current sketch directions. The high-response component is intended to reduce the chance that rare large rows dominating the ℓ_4 response are absent from the memory.

For a later window w , the input state is the carried sketch $B^{(w-1)} = \Sigma_{w-1} V_{w-1}^\top$ and the old row memory $R^{(w-1)}$. The method optimizes the corrected objective $\text{score}_w(v)$ above and again computes orthonormal retained directions sequentially by solving

$$v_j^{(w)} \in \arg \max_{v \in \mathbb{S}^{d-1}, (v_i^{(w)})^\top v = 0 \ \forall i < j} \text{score}_w(v). \quad (29)$$

After selecting the new right space, it converts that space into the left-projected carry sketch. With

$$\hat{V}_w = [v_1^{(w)}, \dots, v_r^{(w)}], \quad M_w := \begin{pmatrix} B^{(w-1)} \\ A_w \end{pmatrix}, \quad (30)$$

form

$$Y_w := M_w \widehat{V}_w, \quad U_{,w} := \text{orth}(Y_w), \quad (31)$$

and compute

$$U_{,w}^\top M_w = U_{\text{small},w} \Sigma_w V_w^\top. \quad (32)$$

The sketch passed to the next window is

$$B^{(w)} = \Sigma_w V_w^\top. \quad (33)$$

The row memory is also updated so that $R^{(w)}$ samples from the raw rows seen through window w , not from the sketch rows.

Derivation of the new-window contribution. Let

$$z = A_w v \quad (34)$$

be the response of the current window, and let

$$x = A v \quad (35)$$

be the response of the full remaining population represented by n_w . Since A_w contains only $k = \text{row}(A_w)$ rows, z is a restriction of x . Use the normalized-entropy heuristic

$$\frac{h_2\left(\frac{x}{\|x\|_2}\right)}{\log n_w} \approx \frac{h_2\left(\frac{z}{\|z\|_2}\right)}{\log k}. \quad (36)$$

Using

$$h_2\left(\frac{y}{\|y\|_2}\right) = 4 \log \|y\|_2 - 4 \log \|y\|_4, \quad (37)$$

and $\|z\|_4 \leq \|x\|_4$, we obtain

$$4 \log \frac{\|z\|_2}{\|x\|_2} \leq h_2\left(\frac{z}{\|z\|_2}\right) - h_2\left(\frac{x}{\|x\|_2}\right). \quad (38)$$

Substituting the normalized-entropy scaling gives

$$\log \frac{\|z\|_2}{\|x\|_2} \lesssim -\frac{h_2\left(\frac{z}{\|z\|_2}\right)}{4} \frac{\log(n_w/k)}{\log k}, \quad (39)$$

hence

$$\frac{\|A_w v\|_2}{\|A v\|_2} \lesssim \left(\frac{k}{n_w}\right)^{\frac{h_2\left(\frac{A_w v}{\|A_w v\|_2}\right)}{4 \log k}}. \quad (40)$$

This motivates the reciprocal entropy factor

$$\left(\frac{k}{n_w}\right)^{-\frac{h_2\left(\frac{A_w v}{\|A_w v\|_2}\right)}{4 \log k}}. \quad (41)$$

Unlike the earlier simplified score, the corrected score does not add the flat-case prefactor $(k/n_w)^{1/4}$. The missing information in the old term is handled directly by the old row-sample entropy proxy, rather than by a fixed rebalance of the new term.

Derivation of the old contribution. The sketch gives the old retained energy

$$E_{\text{old}}(v) = \|Bv\|_2^2. \quad (42)$$

It does not give row-level entropy. The row memory R supplies a proxy

$$h_2^{(R)}(v) = 4 \log \|Rv\|_2 - 4 \log \|Rv\|_4. \quad (43)$$

Under normalized-entropy scaling,

$$h_2^{(\text{old})}(v) \approx h_2^{(R)}(v) \frac{\log n_{\text{seen}}}{\log s}. \quad (44)$$

Applying the same reciprocal entropy correction at the old-row fraction n_{seen}/n gives

$$\psi_{\text{old}}(v) = \|Bv\|_2^2 \left(\frac{n_{\text{seen}}}{n}\right)^{-\frac{h_2^{(R)}(v)}{4 \log s}}. \quad (45)$$

Thus the old term now has the same structure as the new term: retained energy multiplied by a direction-dependent entropy correction.

Simplified form. Define

$$c_w := \frac{\log(k/n_w)}{\log k}, \quad (46)$$

$$c_{\text{old}} := \frac{\log(n_{\text{seen}}/n)}{\log s}. \quad (47)$$

Then

$$\psi_{\text{new}}(v) = \|A_w v\|_2^{2-c_w} \|A_w v\|_4^{c_w}, \quad (48)$$

$$\psi_{\text{old}}(v) = \|Bv\|_2^2 \|Rv\|_2^{-c_{\text{old}}} \|Rv\|_4^{c_{\text{old}}}. \quad (49)$$

Therefore

$$\text{score}_w(v) = \|Bv\|_2^2 \|Rv\|_2^{-c_{\text{old}}} \|Rv\|_4^{c_{\text{old}}} + \|A_w v\|_2^{2-c_w} \|A_w v\|_4^{c_w}. \quad (50)$$

When $k < n_w$ and $n_{\text{seen}} < n$, both c_w and c_{old} are nonpositive, so high-entropy responses are boosted by the same normalized-entropy mechanism in both old and new contributions.

Exact gradient. Let

$$y_B := Bv, \quad y_R := Rv, \quad z := A_w v, \quad (51)$$

and assume the denominators below are nonzero.

For the old contribution,

$$\log \psi_{\text{old}}(v) = \log \|Bv\|_2^2 - c_{\text{old}} \log \|Rv\|_2 + c_{\text{old}} \log \|Rv\|_4. \quad (52)$$

Differentiating gives

$$\nabla \log \psi_{\text{old}}(v) = 2 \frac{B^\top Bv}{\|Bv\|_2^2} - c_{\text{old}} \frac{R^\top Rv}{\|Rv\|_2^2} + c_{\text{old}} R^\top \left(\frac{(Rv)^{\odot 3}}{\|Rv\|_4^4} \right). \quad (53)$$

Hence

$$\nabla \psi_{\text{old}}(v) = \psi_{\text{old}}(v) \left[2 \frac{B^\top Bv}{\|Bv\|_2^2} - c_{\text{old}} \frac{R^\top Rv}{\|Rv\|_2^2} + c_{\text{old}} R^\top \left(\frac{(Rv)^{\odot 3}}{\|Rv\|_4^4} \right) \right]. \quad (54)$$

For the new-window contribution,

$$\log \psi_{\text{new}}(v) = (2 - c_w) \log \|A_w v\|_2 + c_w \log \|A_w v\|_4. \quad (55)$$

Differentiating gives

$$\nabla \log \psi_{\text{new}}(v) = (2 - c_w) \frac{A_w^\top A_w v}{\|A_w v\|_2^2} + c_w A_w^\top \left(\frac{(A_w v)^{\odot 3}}{\|A_w v\|_4^4} \right). \quad (56)$$

Hence

$$\nabla \psi_{\text{new}}(v) = \psi_{\text{new}}(v) \left[(2 - c_w) \frac{A_w^\top A_w v}{\|A_w v\|_2^2} + c_w A_w^\top \left(\frac{(A_w v)^{\odot 3}}{\|A_w v\|_4^4} \right) \right]. \quad (57)$$

Therefore the full Euclidean gradient is

$$\nabla \text{score}_w(v) = \nabla \psi_{\text{old}}(v) + \nabla \psi_{\text{new}}(v). \quad (58)$$

Equivalently,

$$\nabla \text{score}_w(v) = \psi_{\text{old}}(v) \left[2 \frac{B^\top Bv}{\|Bv\|_2^2} - c_{\text{old}} \frac{R^\top Rv}{\|Rv\|_2^2} + c_{\text{old}} R^\top \left(\frac{(Rv)^{\odot 3}}{\|Rv\|_4^4} \right) \right] \quad (59)$$

$$+ \psi_{\text{new}}(v) \left[(2 - c_w) \frac{A_w^\top A_w v}{\|A_w v\|_2^2} + c_w A_w^\top \left(\frac{(A_w v)^{\odot 3}}{\|A_w v\|_4^4} \right) \right]. \quad (60)$$

Expanded old-gradient form. Sometimes it is numerically preferable to expose the quadratic sketch-energy gradient instead of using the logarithmic form. Let

$$\phi_R(v) := \|Rv\|_2^{-c_{\text{old}}} \|Rv\|_4^{c_{\text{old}}}, \quad E_{\text{old}}(v) := \|Bv\|_2^2. \quad (61)$$

Then

$$\psi_{\text{old}}(v) = E_{\text{old}}(v)\phi_R(v), \quad (62)$$

and

$$\nabla\psi_{\text{old}}(v) = 2\phi_R(v)B^\top Bv + E_{\text{old}}(v)\phi_R(v) \left[-c_{\text{old}} \frac{R^\top Rv}{\|Rv\|_2^2} + c_{\text{old}} R^\top \left(\frac{(Rv)^{\odot 3}}{\|Rv\|_4^4} \right) \right]. \quad (63)$$

This form remains meaningful when $\|Bv\|_2$ is zero, provided the old entropy proxy is well-defined.

Stored-basis form. If

$$B = SV^\top, \quad V = [v_1, \dots, v_r], \quad S = \text{diag}(s_1, \dots, s_r), \quad (64)$$

then

$$\|Bv\|_2^2 = \sum_{j=1}^r s_j^2 (v_j^\top v)^2, \quad B^\top Bv = \sum_{j=1}^r s_j^2 (v_j^\top v) v_j. \quad (65)$$

Thus the expanded old-gradient term may be written as

$$\nabla\psi_{\text{old}}(v) = 2\phi_R(v) \sum_{j=1}^r s_j^2 (v_j^\top v) v_j + \psi_{\text{old}}(v) \left[-c_{\text{old}} \frac{R^\top Rv}{\|Rv\|_2^2} + c_{\text{old}} R^\top \left(\frac{(Rv)^{\odot 3}}{\|Rv\|_4^4} \right) \right]. \quad (66)$$

The full gradient is this old term plus $\nabla\psi_{\text{new}}(v)$.

Optimization problem. To extract a direction, solve

$$\max_{v \in \mathbb{S}^{d-1}, Q^\top v = 0} \text{score}_w(v), \quad (67)$$

where the columns of Q are the previously accepted directions.

Projected full-gradient ascent. Use the exact gradient of the corrected score. Project it onto the feasible tangent space:

$$g_{\text{tan}}(v) := (I - QQ^\top - vv^\top) \nabla \text{score}_w(v). \quad (68)$$

Then take

$$\tilde{v}^+ = v + \eta g_{\text{tan}}(v), \quad v^+ = \frac{(I - QQ^\top) \tilde{v}^+}{\|(I - QQ^\top) \tilde{v}^+\|_2}, \quad (69)$$

with $\eta > 0$ chosen by backtracking line search on the true corrected score.

Basis-expansion interpretation. If one is building a local search subspace rather than taking a pure steepest-ascent step, the new direction added to the basis is the projected full gradient

$$g_{\text{basis}}(v) := (I - QQ^\top - vv^\top) \nabla \text{score}_w(v). \quad (70)$$

Thus the basis is enlarged by the actual gradient direction of the corrected score. The roles of old sketch energy, old sampled entropy, current-window energy, and current-window entropy are already combined inside this single full-gradient vector.

In a reduced-subspace version, suppose the current local right search basis is

$$Z \in \mathbb{R}^{d \times q}, \quad Z^\top Z = I. \quad (71)$$

After solving the score problem inside $\text{span}(Z)$, the relevant expansion residual is the part of the feasible tangent gradient that is not already represented by this local basis:

$$r_Z(v) := (I - ZZ^\top) (I - QQ^\top - vv^\top) \nabla \text{score}_w(v). \quad (72)$$

The local basis should be expanded only when this residual is non-negligible. A natural stopping rule is

$$\frac{\|r_Z(v)\|_2}{\|(I - QQ^\top - vv^\top) \nabla \text{score}_w(v)\|_2} \leq \tau, \quad (73)$$

where τ is a prescribed tolerance. If the ratio is larger than τ , the new basis direction should be the normalized residual $r_Z(v)/\|r_Z(v)\|_2$, optionally followed by Krylov refinements generated from this residual direction.

Operator extraction from the retained right subspace. The implementation does not stop after constructing a right search space. After the scored directions and any auxiliary directions are assembled into an orthonormal right basis

$$V \in \mathbb{R}^{d \times r}, \quad (74)$$

with row-wise representation $V^\top = V_t$, it forms

$$M_w := \begin{pmatrix} B \\ A_w \end{pmatrix}, \quad Y := M_w V. \quad (75)$$

It then orthonormalizes the image:

$$U := \text{orth}(Y) = \text{orth}(M_w V). \quad (76)$$

Equivalently, this defines the orthogonal projector onto the left image of the retained right space,

$$P_U := UU^\top. \quad (77)$$

The algorithm then compresses M_w to this left subspace:

$$\widetilde{M} := U^\top M_w, \quad (78)$$

which is the coordinate representation of $P_U M_w$. Finally it computes

$$\widetilde{M} = U_{\text{small}} \Sigma V_{\text{new}}^\top, \quad (79)$$

and stores

$$U_{\text{proj}} = U U_{\text{small}}, \quad S = \Sigma, \quad V^\top = V_{\text{new}}^\top. \quad (80)$$

So the carried singular values and right singular vectors are not merely the raw scored-plus-auxiliary basis; they are the singular factors of the left-projected operator $P_U M_w$ restricted to the retained right span.

Why this is the sketch carried into the next window. The next window does not receive the raw optimized right basis V by itself. Instead it receives the sketch

$$B_{\text{next}} := \Sigma V_{\text{new}}^\top, \quad (81)$$

where

$$U^\top M_w = U_{\text{small}} \Sigma V_{\text{new}}^\top \quad (82)$$

is the SVD of the compressed left-projected operator. Equivalently, B_{next} is the norm-preserving right-action representation of

$$P_U M_w = U U^\top M_w. \quad (83)$$

For any direction v ,

$$\|B_{\text{next}} v\|_2 = \|\Sigma V_{\text{new}}^\top v\|_2 = \|P_U M_w v\|_2. \quad (84)$$

Thus the quadratic energy used in the next old term is exactly the energy of the retained compressed operator. The additional row memory R_{next} does not replace this sketch; it supplies only the missing row-level entropy proxy for the next old contribution.

First-window special case. If $B = 0$ and $R = \emptyset$, then

$$c_1 := \frac{\log(k/n_1)}{\log k}, \quad (85)$$

and

$$\text{score}_1(v) = \|A_1 v\|_2^{2-c_1} \|A_1 v\|_4^{c_1}. \quad (86)$$

The full gradient reduces to

$$\nabla \text{score}_1(v) = \|A_1 v\|_2^{2-c_1} \|A_1 v\|_4^{c_1} \left[(2 - c_1) \frac{A_1^\top A_1 v}{\|A_1 v\|_2^2} + c_1 A_1^\top \left(\frac{(A_1 v)^{\odot 3}}{\|A_1 v\|_4^4} \right) \right]. \quad (87)$$

There is no $(k/n_1)^{1/4}$ prefactor.

Practical remarks. The objective is nonconvex, so multiple restarts are useful. One should reject or regularize steps when any of $\|A_w v\|_2$, $\|A_w v\|_4$, $\|Rv\|_2$, $\|Rv\|_4$, or $\|Bv\|_2$ is numerically tiny in a formula that divides by that quantity. The expanded old-gradient form avoids division by $\|Bv\|_2^2$, but the row-sample entropy derivatives still require nontrivial response on the sample.

The row memory should not be interpreted as a replacement for the old sketch. The sketch B supplies the retained old energy, while R supplies only a direction-dependent entropy proxy. A pure reservoir sample gives coverage, but it can miss rare large rows that dominate ℓ_4 terms. A practical memory should therefore combine a uniform reservoir with a small high-response set selected using current sketch directions.